

PUBLISHED: 31-AUG-2017
2014.0 RANGE ROVER (LG), 303-01

ENGINE - V8 S/C 5.0L PETROL

LEFT CYLINDER HEAD (G1509344)

REMOVAL AND INSTALLATION

12.29.04	CYLINDER HEAD - ASSEMBLY - LEFT /ONE ONLY - RENEW	5000 CC, AJ V8, SUPERCHARGED	18.9	USED WITHINS	
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SPECIAL TOOL(S)

 E129632		303-1492 Guide Sleeve, Cylinder Head Bolt
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PART(S)

STEP	PART NAME	QUANTITY
Installation Step 2	Cylinder head gasket	1
Installation Step 3	Cylinder head bolts	1
Installation Step 13	Cylinder block coolant outlet O-ring seals	1

REMOVAL

NOTES:

- Removal steps in this procedure may contain installation details.
- Some variation in the illustrations may occur, but the essential information is always correct.

1. Disconnect the battery ground cable.

Refer to: Specifications (414-00 Battery and Charging System - General Information, Specifications).

2.



WARNING:

Make sure to support the vehicle with axle stands.

Raise and support the vehicle.

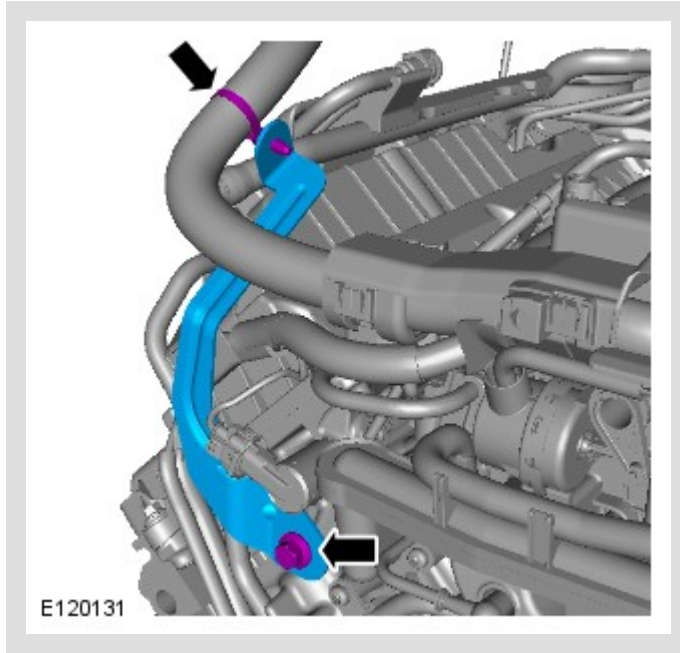
3.

Refer to: Left Camshaft (303-01 Engine - V8 S/C 5.0L Petrol, Removal and Installation).

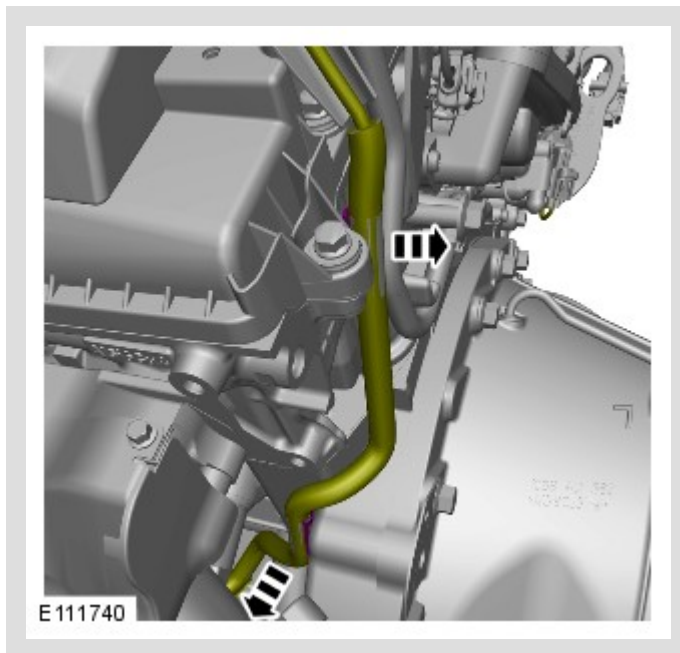
4.

Refer to: Left Exhaust Manifold (303-01 Engine - V8 S/C 5.0L Petrol, Removal and Installation).

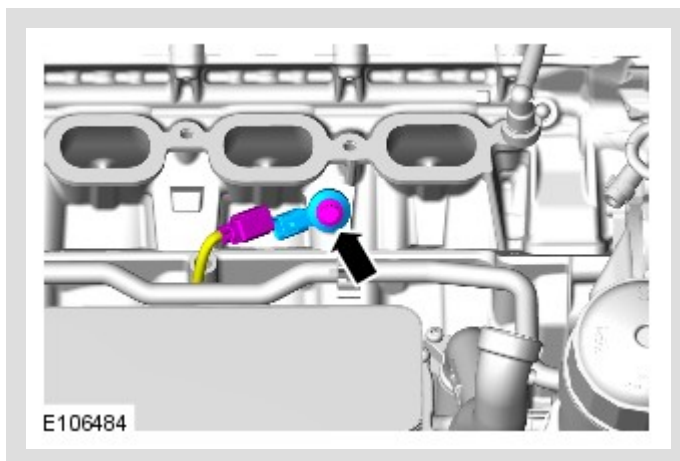
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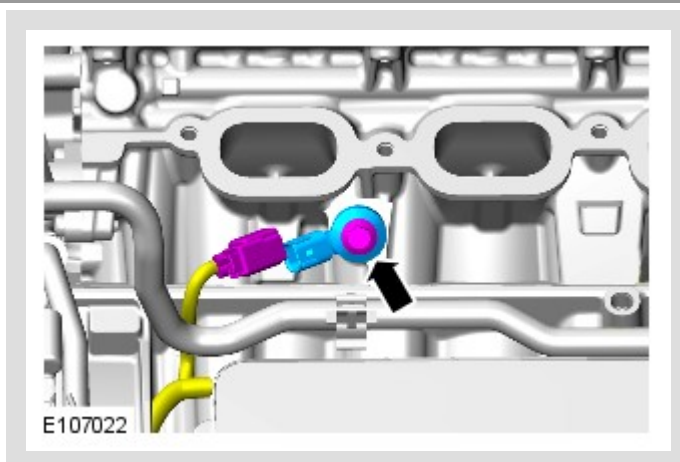
6.



7.



8.

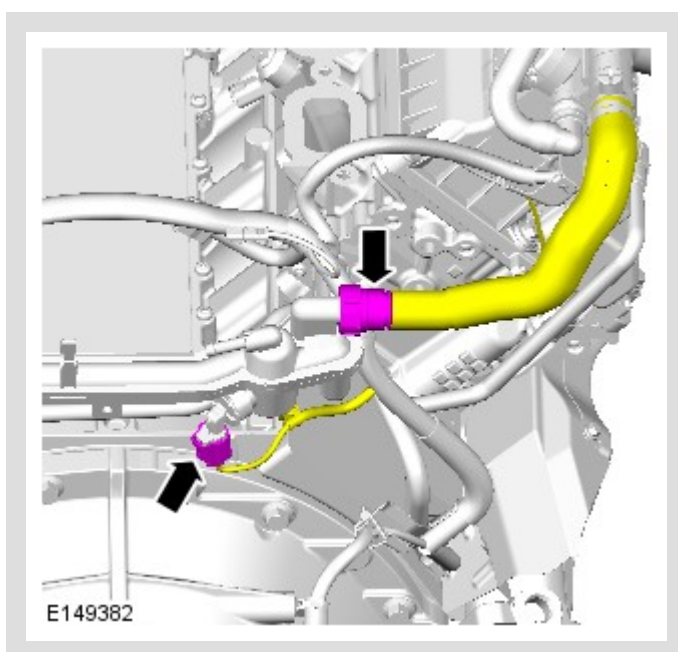


9.



CAUTION:

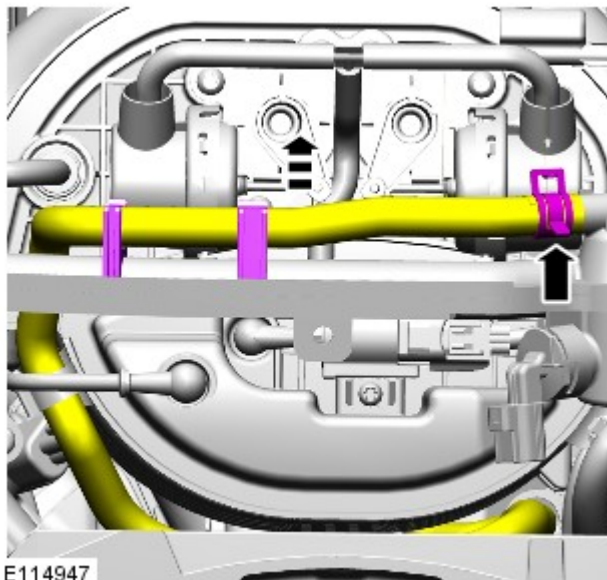
Be prepared to collect escaping coolant.



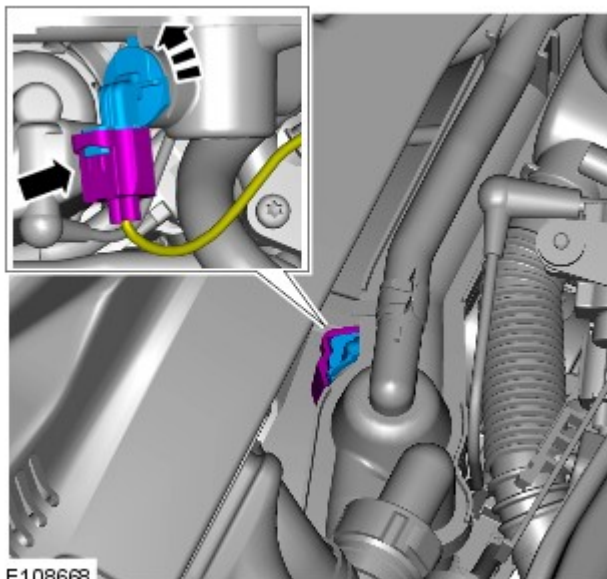
10.

 NOTE:

Clamp the hose to minimize coolant loss.



11.



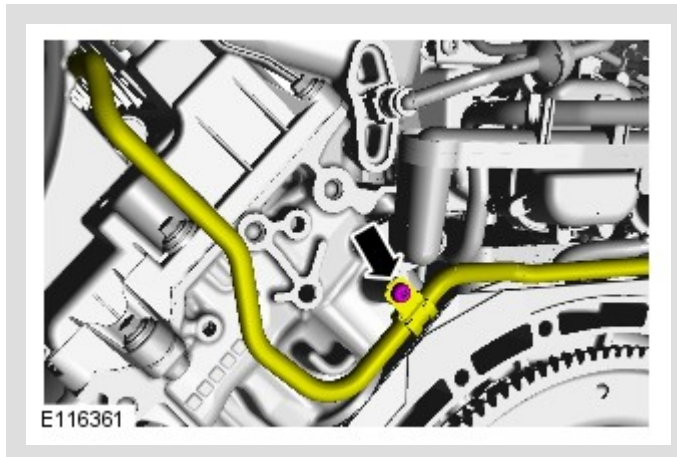
12.

⚠ CAUTION:

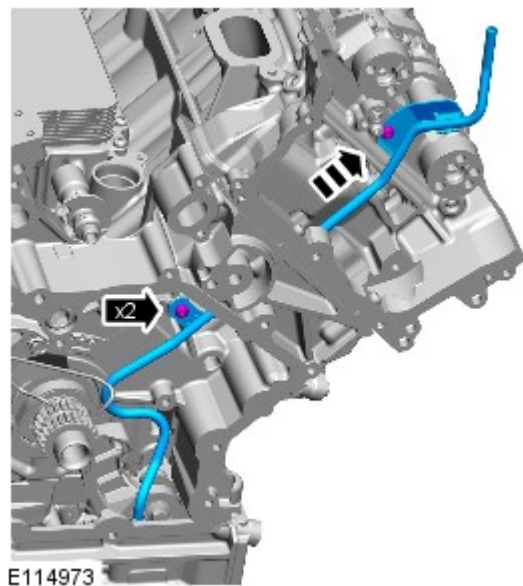
Discard the seals.



13.



14.



15.



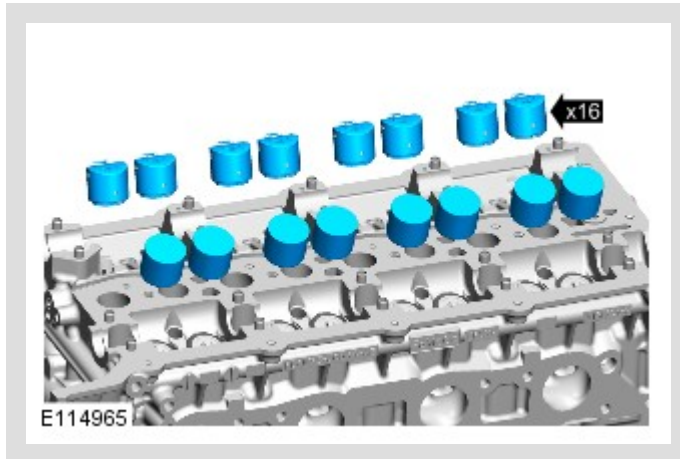
CAUTIONS:

- If a new cylinder head has been installed then new tappets must be installed.
- If the cylinder head is being removed without a new component being installed, the tappets must be installed in their original positions.



NOTE:

Left-hand shown, right-hand similar.

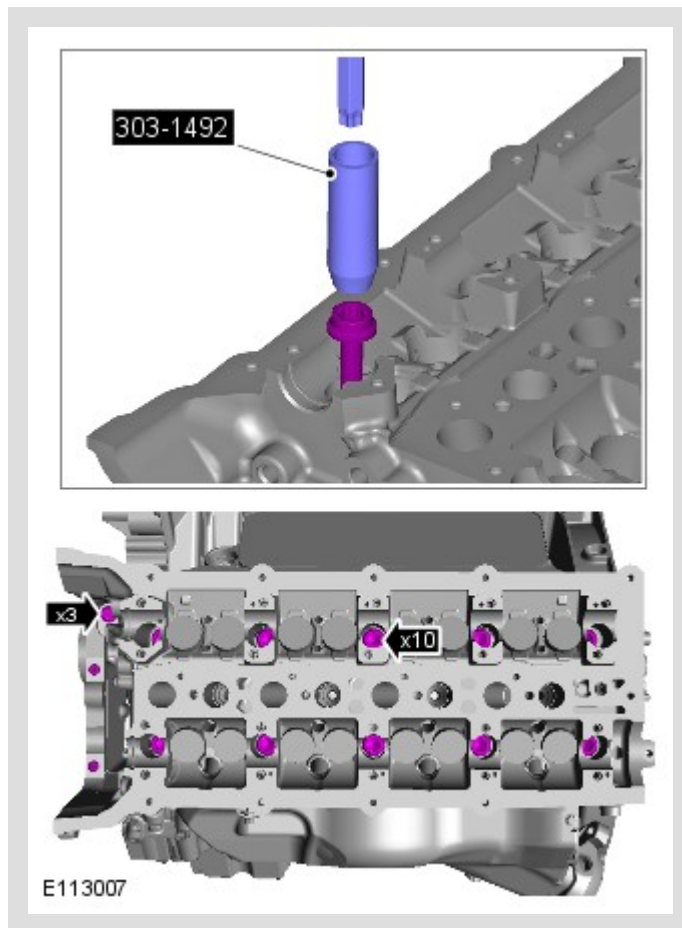


16.

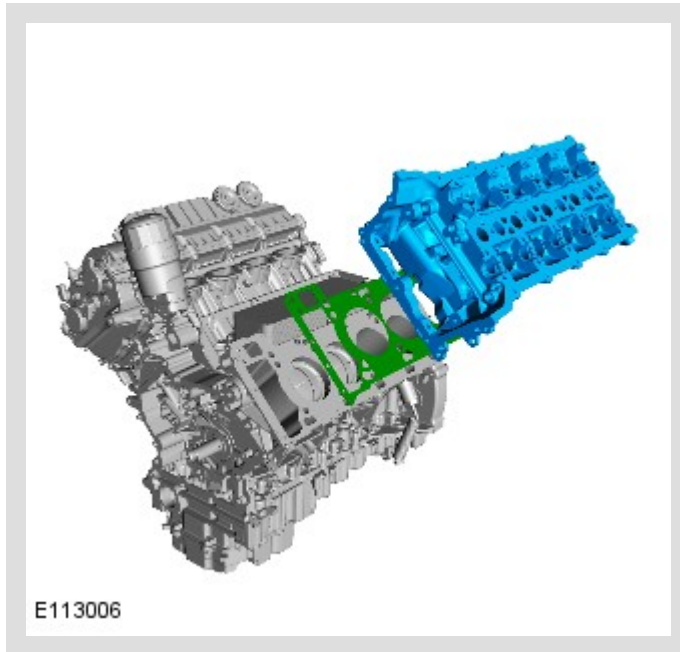


CAUTION:

Discard the bolts.



- *Special Tool(s):* [303-1492](#)



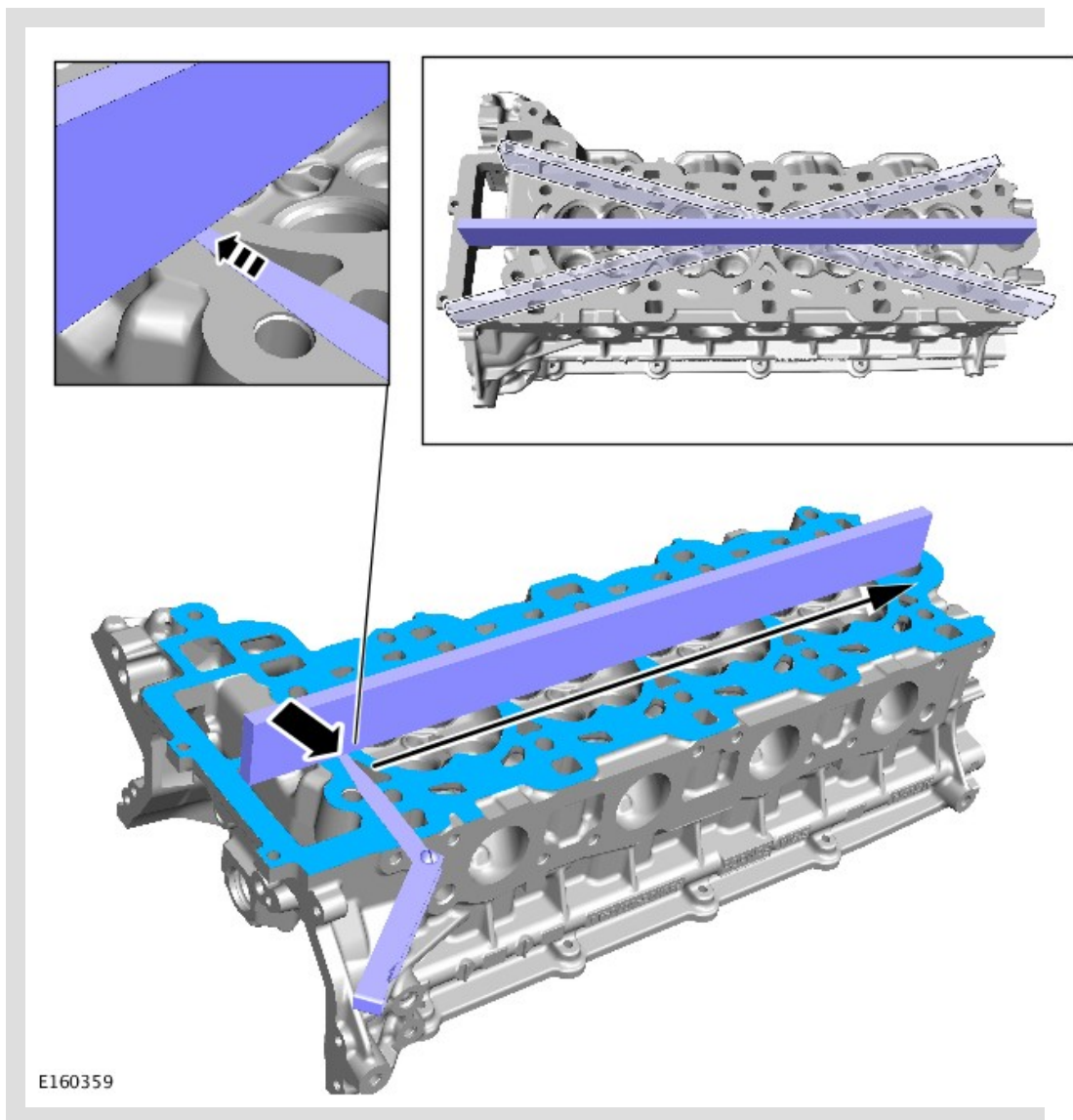
INSTALLATION

1.



CAUTIONS:

- Make sure that the mating faces are clean and free of foreign material.
- An acceptable flatness of the cylinder head is 0.2mm.



- Clean and inspect the cylinder head and cylinder block.
 - Check the cylinder head face for distortion, across the center and from corner to corner.
- Refer to: Cylinder Head Distortion (303-00 Engine System - General Information, General Procedures).



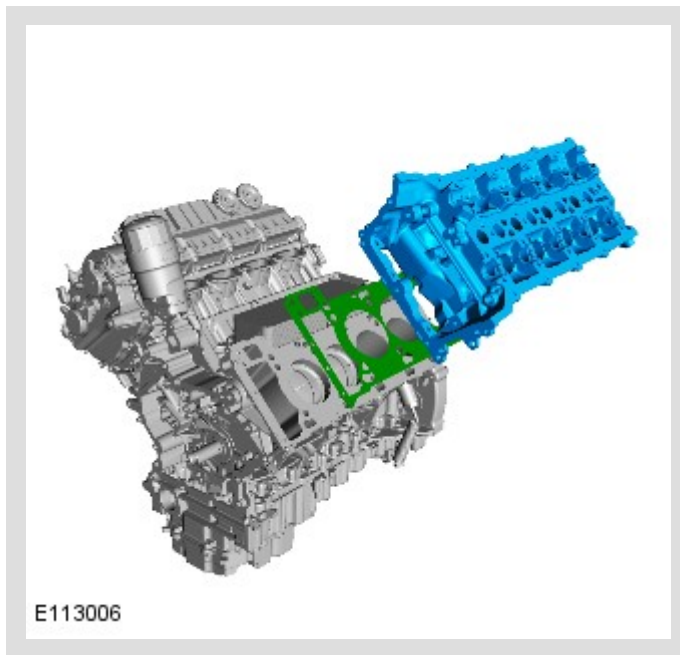
Make sure care is taken when handling the cylinder head gasket.

⚠ CAUTIONS:

- The head gasket must be installed over the cylinder block dowels.
- Make sure that the mating faces are clean and free of foreign material.

△ NOTE:

Install a new cylinder head gasket.



Renew Part: *Cylinder head gasket* Quantity: 1 .

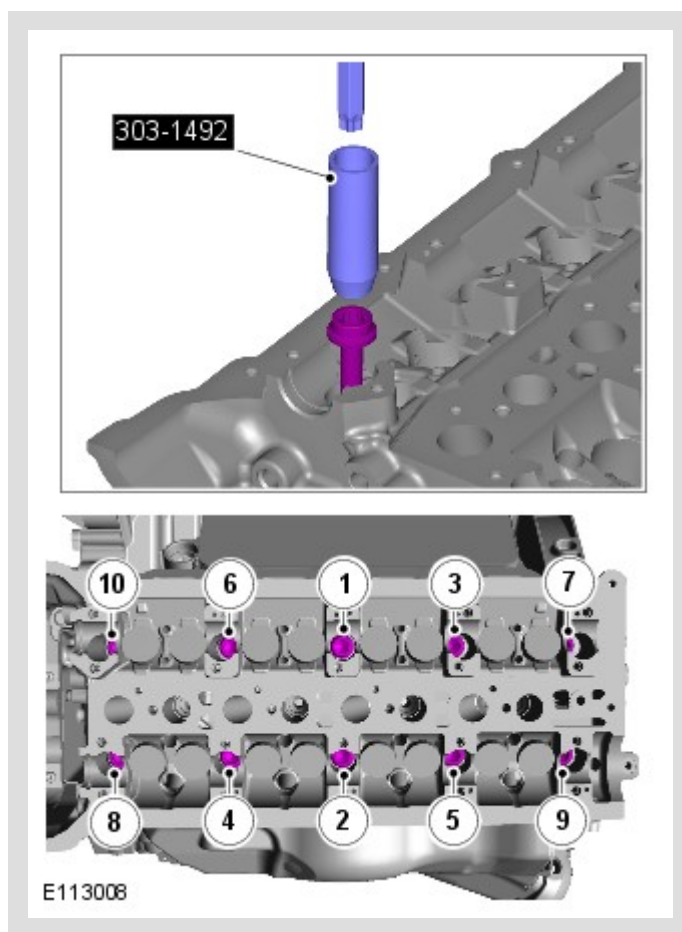
3.

⚠ CAUTION:

Make sure that new cylinder head bolts are installed.

△ NOTE:

Tighten the bolts in the indicated sequence.

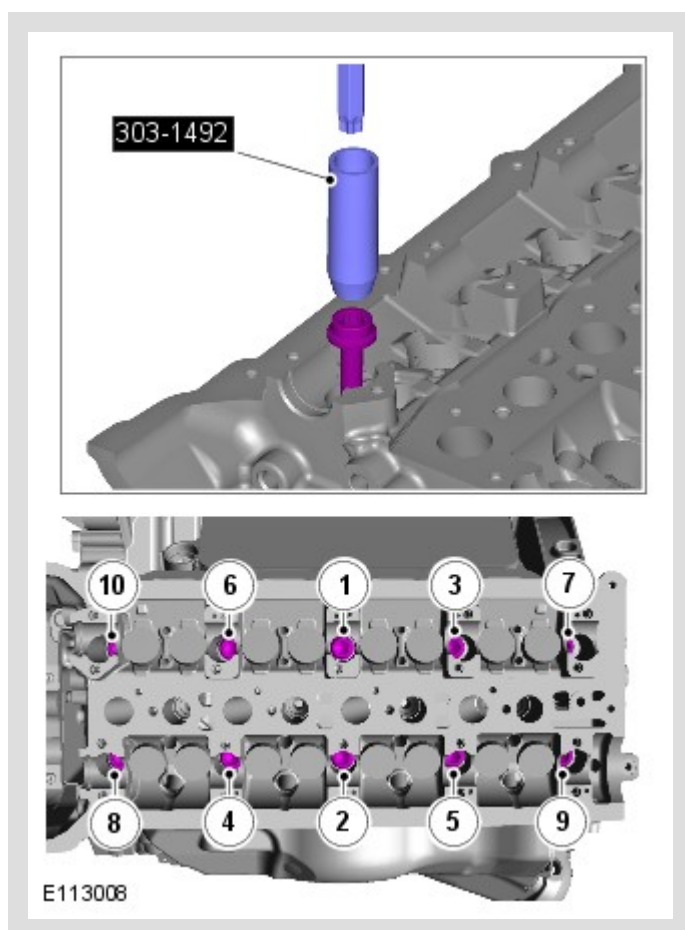


Renew Part: *Cylinder head bolts* Quantity: *1* .
Torque: 20 Nm

4.

 NOTE:

Tighten the bolts in the indicated sequence.

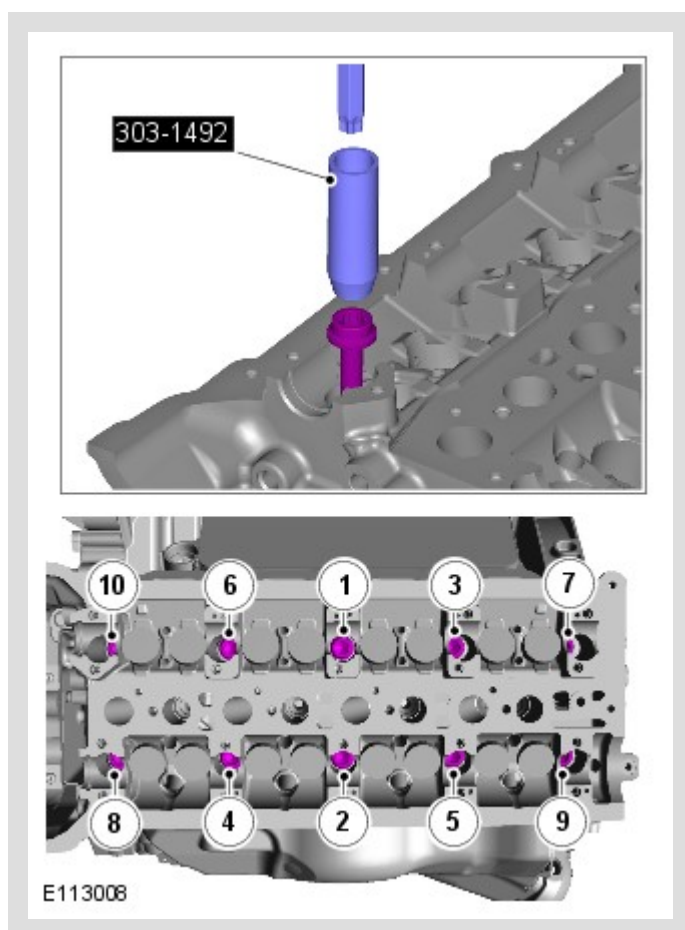


Torque: 35 Nm

5.

 **NOTE:**

Tighten the bolts in the indicated sequence.

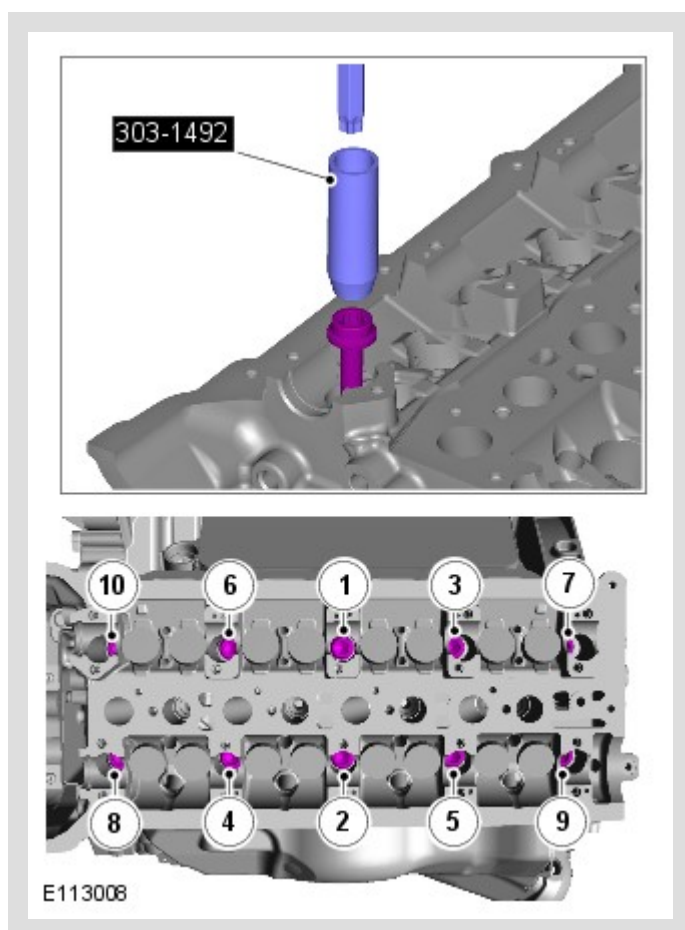


Tighten the bolts 1 to 10, a further 90 degrees.

6.

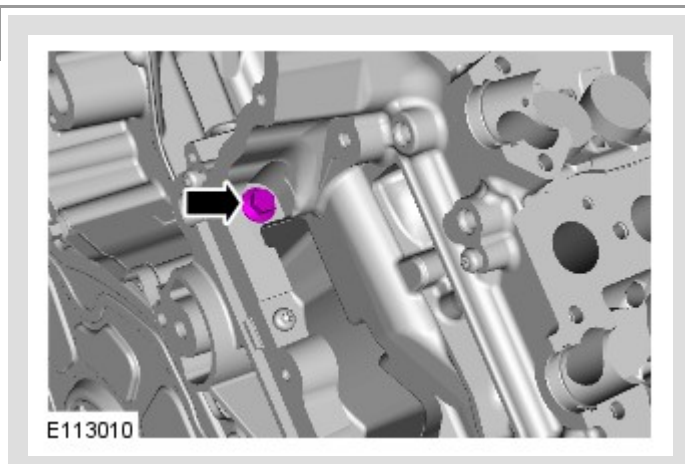
 **NOTE:**

Tighten the bolts in the indicated sequence.



Tighten the bolts 1 to 10, a further 120 degrees.

7.

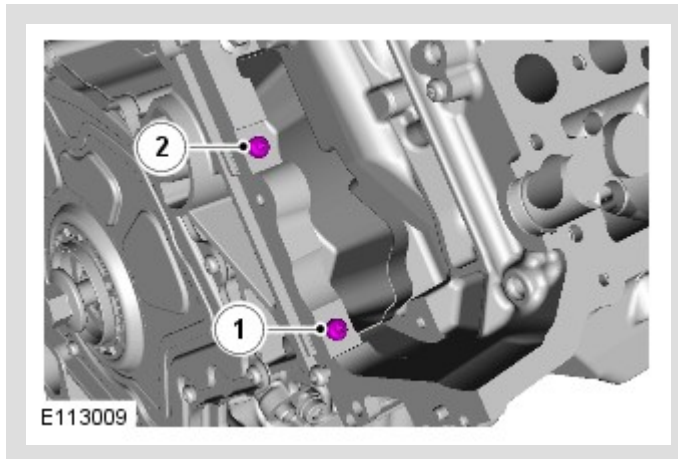


Torque: 25 Nm

8.

 NOTE:

Tighten the bolts in the indicated sequence.

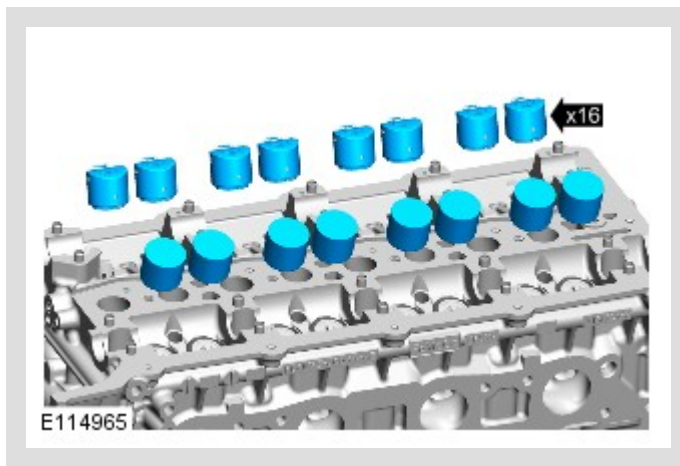


Torque: 12 Nm

9.

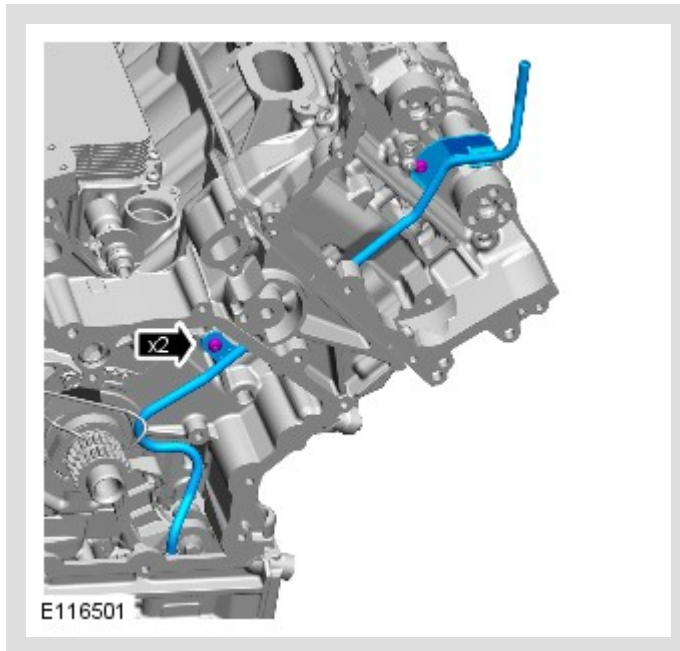
 CAUTIONS:

- If a new cylinder head has been installed then new tappets must be installed.
- If the cylinder head is being removed without a new component being installed, the tappets must be installed in their original positions.



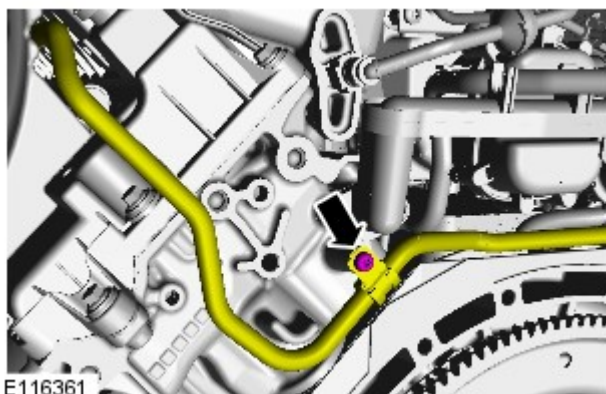
Lubricate the valve tappets with clean engine oil.

10.



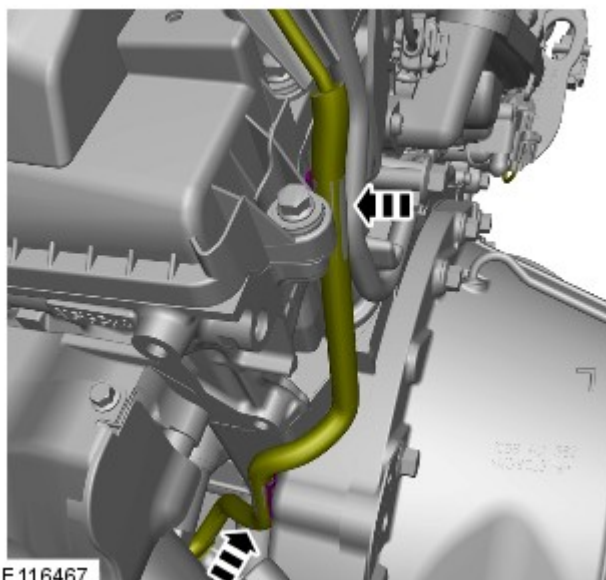
Torque: 12 Nm

11.



Torque: 10 Nm

12.

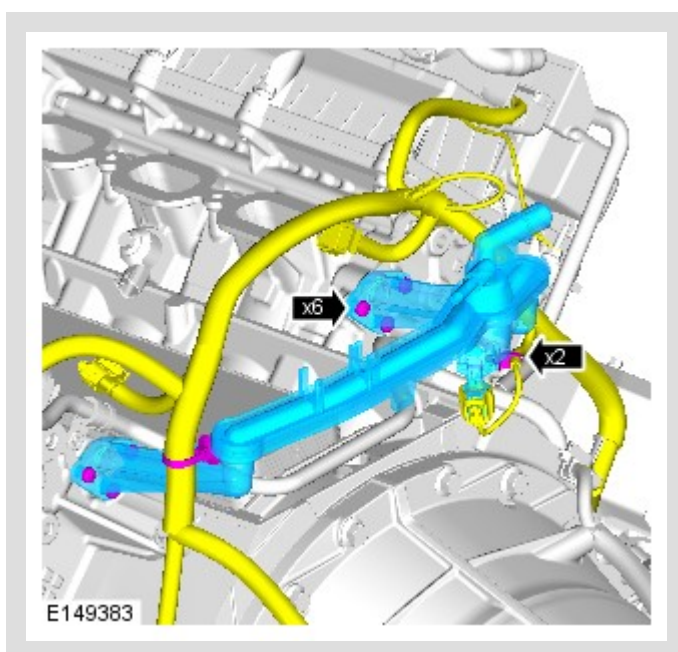


13.



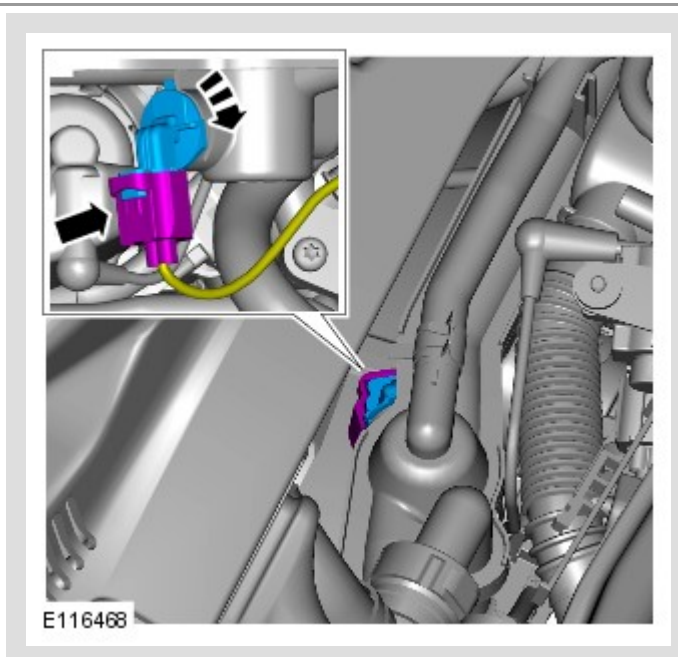
CAUTION:

Install the new seals.

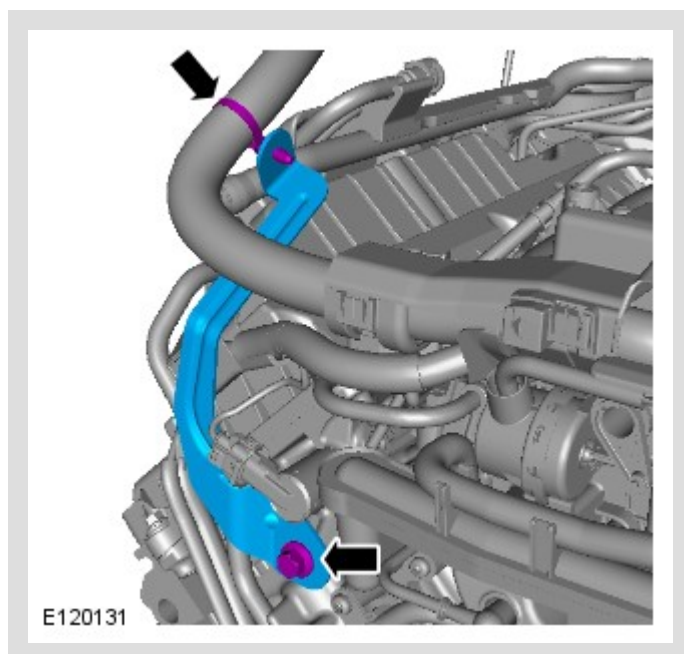


Renew Part: *Cylinder block coolant outlet O-ring seals* Quantity: 1 .
Torque: 10 Nm

14.

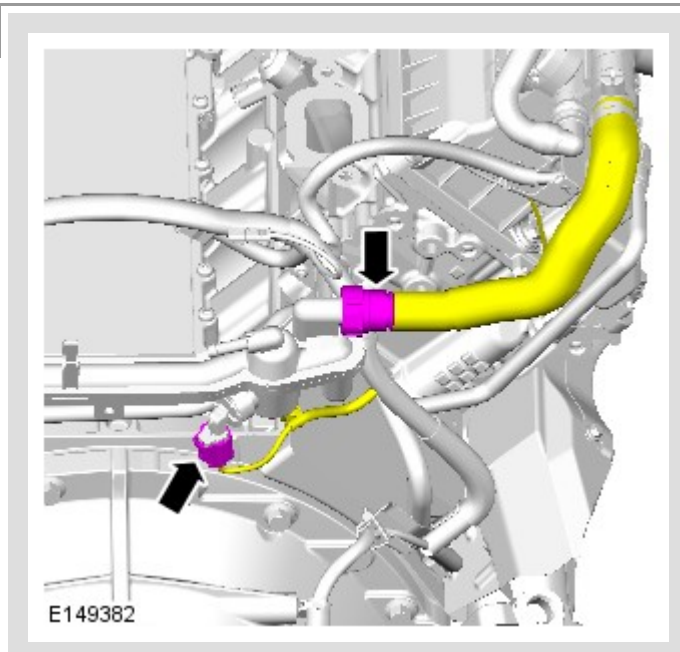


15.

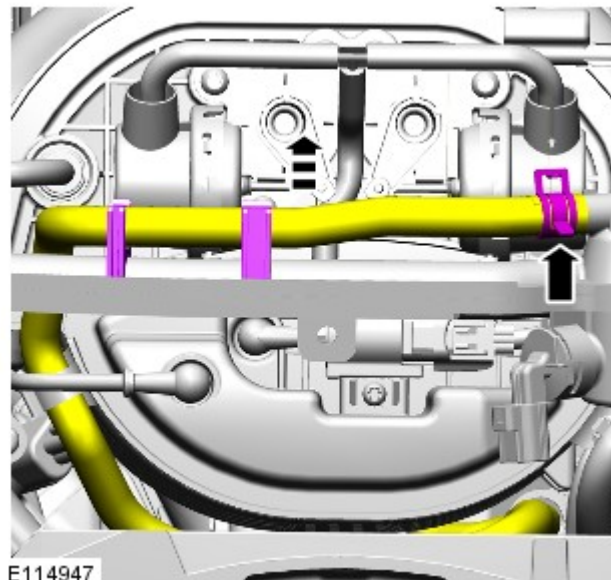


Torque: 25 Nm

16.

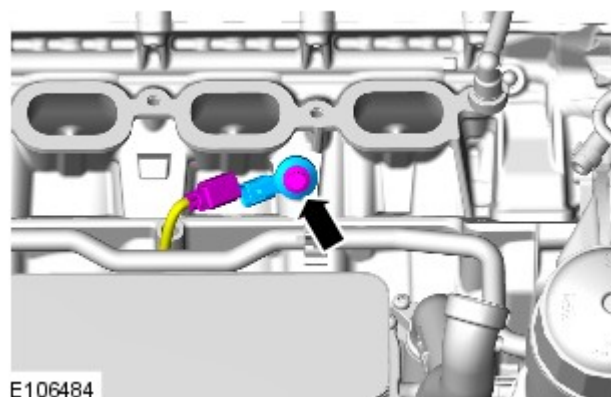


17.



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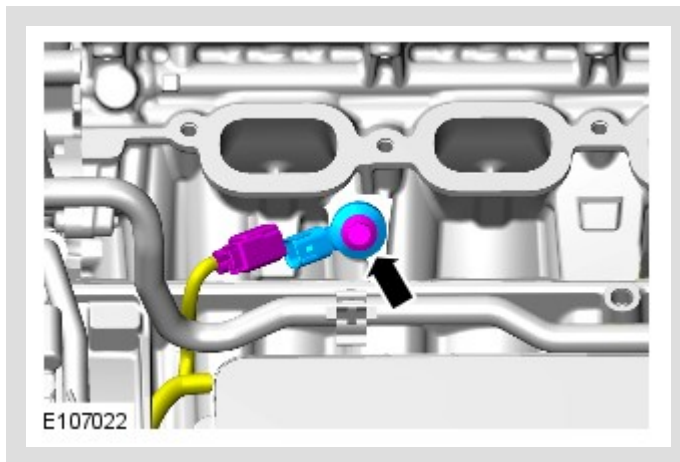
18.



E106484

Torque: 20 Nm

19.



Torque: 20 Nm

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20. Refer to: Left Exhaust Manifold (303-01 Engine - V8 S/C 5.0L Petrol, Removal and Installation).
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21. Refer to: Left Camshaft (303-01 Engine - V8 S/C 5.0L Petrol, Removal and Installation).
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22. Connect the battery ground cable.
Refer to: Specifications (414-00 Battery and Charging System - General Information, Specifications).